

① Encrypt Rail Fence

PT = "Enjoy your Journey" $d = 2, 3$

N O Y U J U n Y

$$d = 3$$

E Y R n x
N O U T U n y
J O O e

CT = EYRR x NOY UJUNYJOOE

$d=2$ P I T C E N

R N S R E

CT: PITCENRNSRE

$d=3$

P T E

R N S R E

I C N

CI = PTERNSREICH

$$d \geq L$$

③ Decrypt :- EOC SNYUWLEJYREIIONS

E	.	.		O															S
	N			Y		U			W			PL							E
		F		X			R		E				A						S
			O					N							S				

PT = ~~RE~~ ENJOY YOUR NEW CLASSES

4. Decrypt BOOIGUYUBOS^NUUTYRKAS d=3

B			O			O			J			G		
	U		Y		U	B	O	S	N	U		U		T
		Y			R			K		A			S	

PT = BUY YOUR BOOKS IN AUGUST

5. SEMSERTISINCMUO

S E M S E R T I

S I N C M U O

PT = SSEIMNSCEMRUTOI

Permutation:

(1) message :- "we are discovered" key : 53124

5 3 1 2 4

W E A R E

D I S C O

V E R E D

CT = ~~W~~ASRRCEEIEEODWDV

DES

(1) key in hexadecimal : 0x0002 0000 0000 0001
Initial permutation is given (table)

0000 0000 0000 0010

0000 0000 0000 0000 bit position

0000 0000 0000 0000 15 -> 63

0000 0000 0000 0001 64 -> 25

Apply initial permutation

0000 0000 0000 0000 0000 0000 1000 0000

0000 0000 0000 0000 0000 0000 0000 0010

Hexa :- 0x000 0008 0000 0001

② 4 bit PT :- $M = 1101$ $K = 1011$ $\rightarrow$ XOR

$$\begin{array}{r} 1101 \\ 1011 \\ \hline 0110 \end{array} \quad CT = 0110$$

③ $1/p \rightarrow$ Sbox 1 = 000000

$$\begin{array}{c} \overbrace{000000} \\ C = 0 \quad \gamma = 0 \\ o/p = 01110 \end{array}$$